Uncertainty Quantification — Concepts

The problem

Language models produce fluent, confident-sounding text whether or not they know the answer. Uncertainty quantification asks a narrower, answerable question: can we compute a number, at generation time, that ranks wrong answers above right ones?

Two granularities

```
TOKEN-LEVEL    one greedy decode, per-step next-token distribution p_t
  entropy      H(p_t)  = -SUM_v p_t(v) log p_t(v)            [nats]
  max-softmax  max_v p_t(v)          margin  top1 - top2
  log-perplexity  -(1/T) SUM_t log p_t(y_t)

SENTENCE-LEVEL  N stochastic samples, compare their MEANINGS
  semantic entropy   H over meaning-cluster probabilities
  lexical entropy    H over exact-match strings   (the ablation)
  self-consistency   share of samples equal to the modal answer
```

Semantic entropy — the key idea

Sampling the same question ten times may give ten different strings that all mean the same thing. Counting strings overstates uncertainty. Kuhn et al. (2023) cluster samples by bidirectional NLI entailment: answers a and b share a meaning class when a entails b AND b entails a. Entropy is then taken over cluster probabilities, not string frequencies.

```
  Discrete   SE = -SUM_c (n_c/N) log(n_c/N)        cluster counts
  Weighted   SE = -SUM_c P(c) log P(c),  P(c) ∝ SUM_{i∈c} exp(logp_i)

  Two traps:  premise must include the QUESTION
              read the entailment index from config.id2label
              (cross-encoders order [contradiction, entailment, neutral])
```

Evaluation metrics

```
AUROC   P(uncertainty of a wrong answer > that of a right one).
        Rank-based (Mann-Whitney U), ties get average ranks.
        0.5 = useless.  Undefined if all-correct or all-wrong.

ECE     SUM_b (n_b/N) |acc(b) - conf(b)|.  Needs a PROBABILITY, so
        raw entropies are mapped through a dev-fitted calibrator.
        Fixed-width bins go empty at small N -> also use equal-mass.

AURC    mean risk over all coverage levels, sorting by confidence.
        Lower is better. Directly models answer-or-abstain.
```

The leakage rule

Thresholds, calibrators and operating points are fitted on the dev split and reported on the held-out test split. Fitting on the split you report makes every method look better calibrated and every abstention threshold look more transferable than it is.

This Run — nq_open

Signal ranking

greedy accuracy 0.140   dev 100 / test 100

signal	cost	AUROC	95% CI
Sequence log-prob	single_pass	0.945	[0.899, 0.981]
Sequence probability	single_pass	0.945	[0.899, 0.981]
Lexical entropy	multi_sample	0.916	[0.848, 0.968]
Self-consistency	multi_sample	0.914	[0.842, 0.966]
Distinct answer strings	multi_sample	0.914	[0.846, 0.965]
Log-perplexity	single_pass	0.914	[0.855, 0.964]
Min max-softmax	single_pass	0.914	[0.849, 0.962]
Distinct meanings	multi_sample	0.907	[0.841, 0.958]

Selective answering

```
dev-fitted thresholds applied to held-out data:
  target 25% -> coverage 10.0%, accuracy 0.600
  target 40% -> coverage  7.0%, accuracy 0.714
  target 50% -> abstained on everything
  target 60% -> abstained on everything
```

This Run — gsm8k

Signal ranking

```
greedy accuracy 0.560   dev 50 / test 50

signal                                cost            AUROC           95% CI
--------------------------------------------------------------------------------
Lexical entropy                       multi_sample    0.884  [0.780, 0.966]
Self-consistency                      multi_sample    0.884  [0.789, 0.965]
Distinct answer strings               multi_sample    0.874  [0.770, 0.961]
Sequence log-prob                     single_pass     0.828  [0.684, 0.945]
Sequence probability                  single_pass     0.828  [0.684, 0.945]
Semantic entropy (likelihood-weighted multi_sample    0.806  [0.674, 0.926]
Semantic self-consistency             multi_sample    0.800  [0.661, 0.918]
Semantic entropy                      multi_sample    0.794  [0.659, 0.911]
```

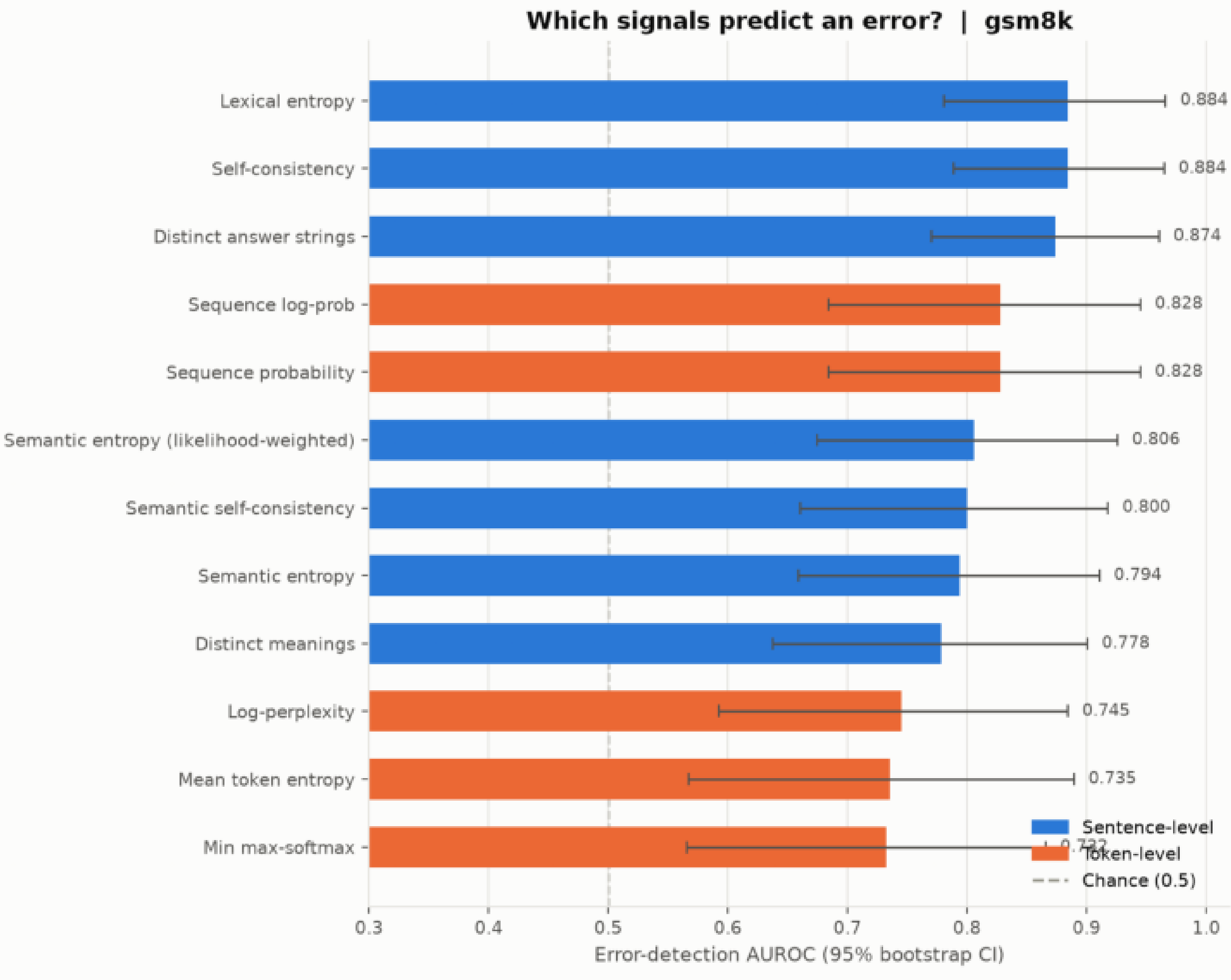
Selective answering

```
dev-fitted thresholds applied to held-out data:
  target 60% -> coverage 100.0%, accuracy 0.560
  target 70% -> coverage 74.0%, accuracy 0.730
  target 80% -> coverage 70.0%, accuracy 0.771
  target 90% -> coverage 44.0%, accuracy 0.864
```

Self-consistency cascade

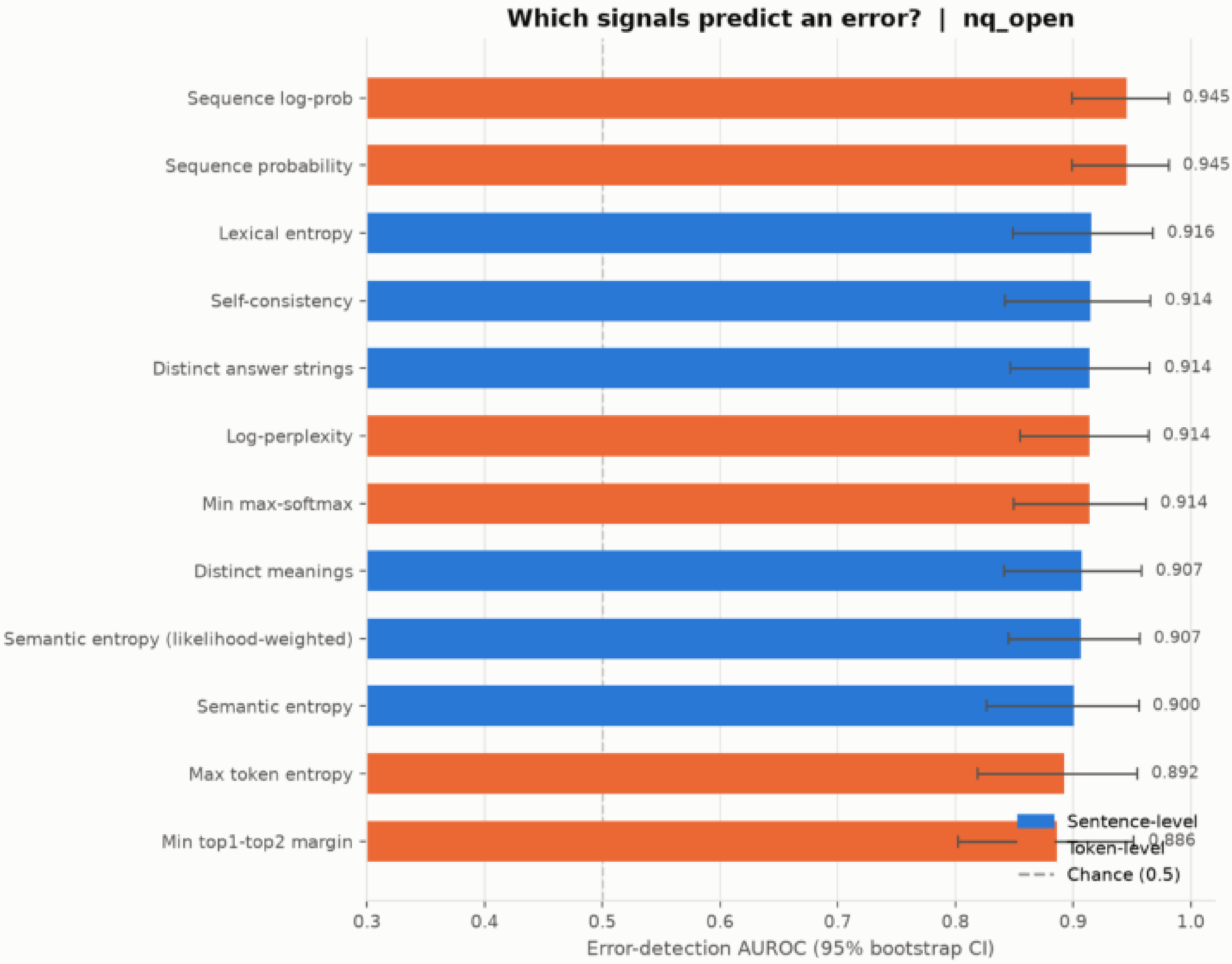
```
  greedy            0.560   1.00 gen
  always sample     0.660  11.00 gen
  gated cascade     0.640   7.80 gen  (68% escalated)
```

Which signals predict an error? — gsm8k



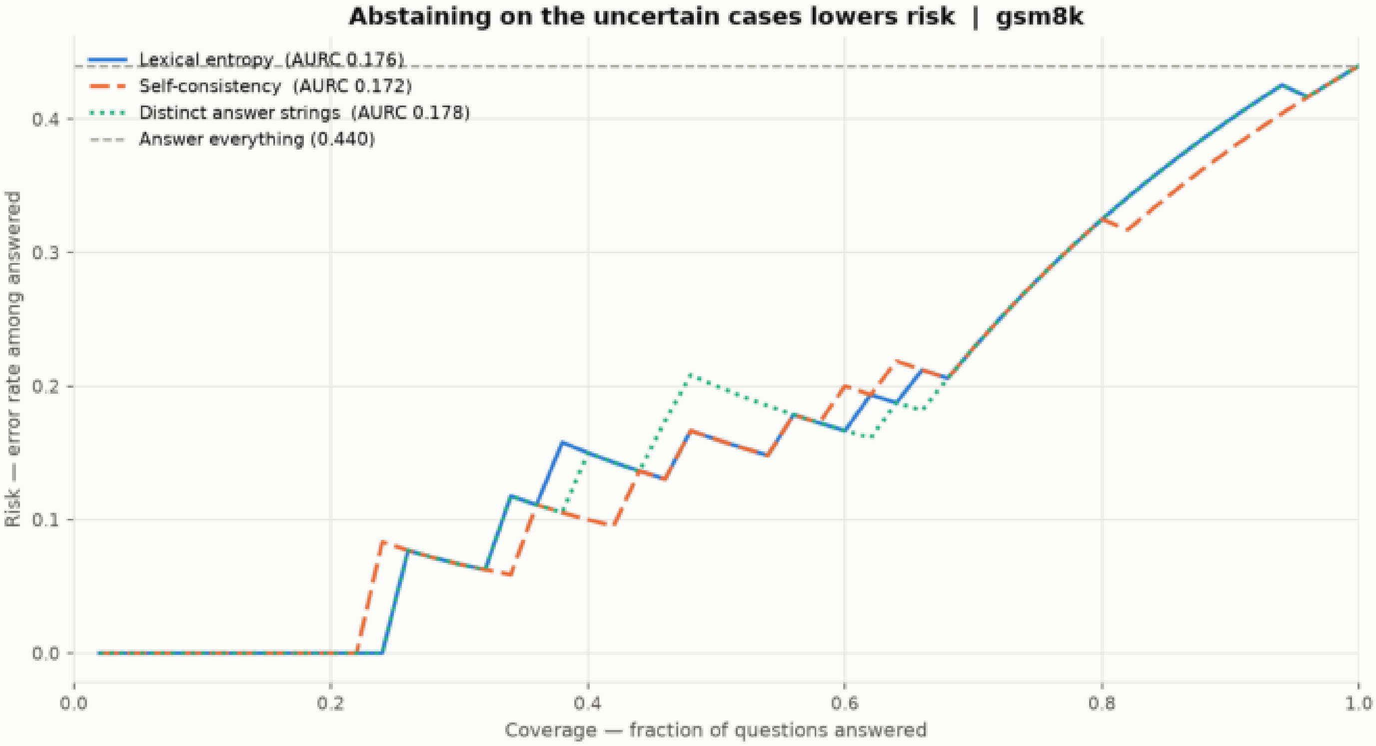
`auroc_by_signal_gsm8k.png`

Which signals predict an error? — nq_open



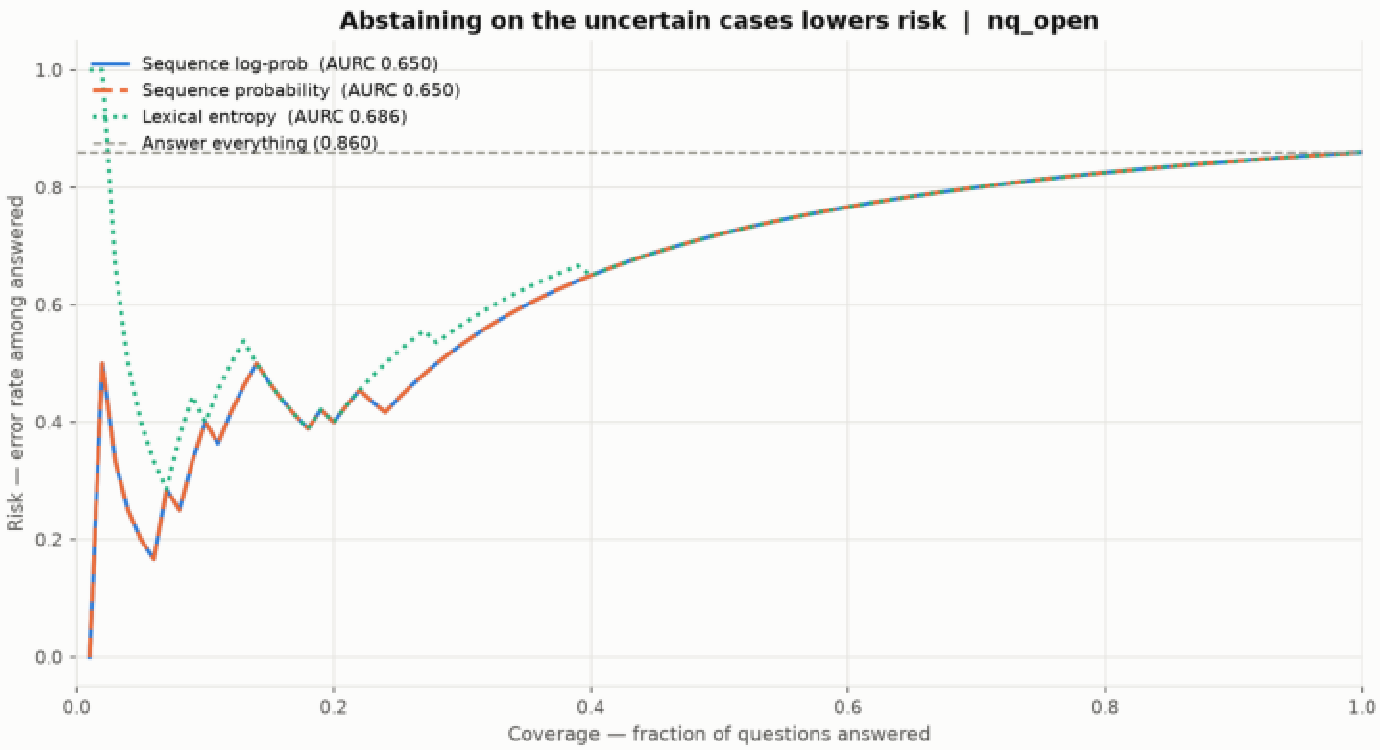
auroc_by_signal_nq_open.png

Risk falls as you abstain on the uncertain cases — gsm8k



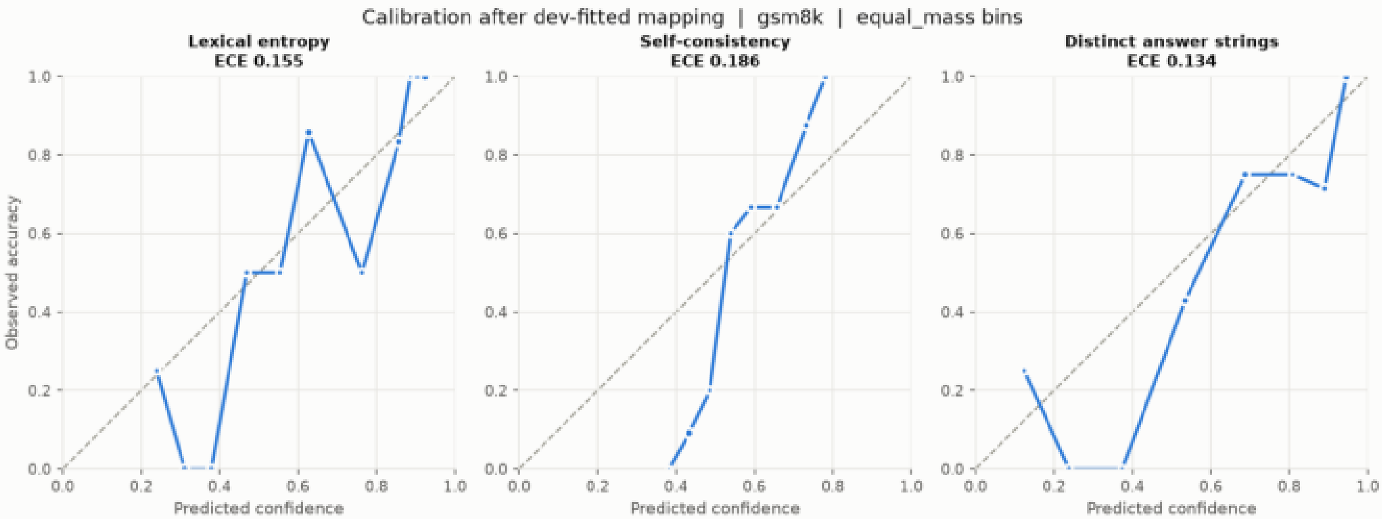
risk_coverage_gsm8k.png

Risk falls as you abstain on the uncertain cases — nq_open



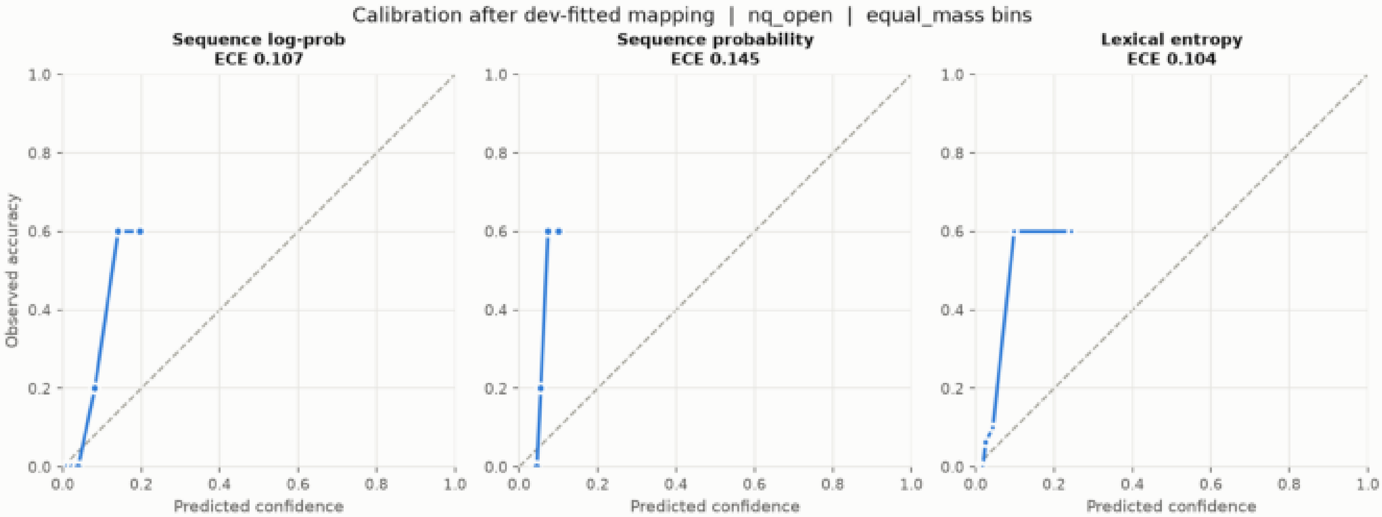
risk_coverage_nq_open.png

Calibration, equal-mass bins — gsm8k



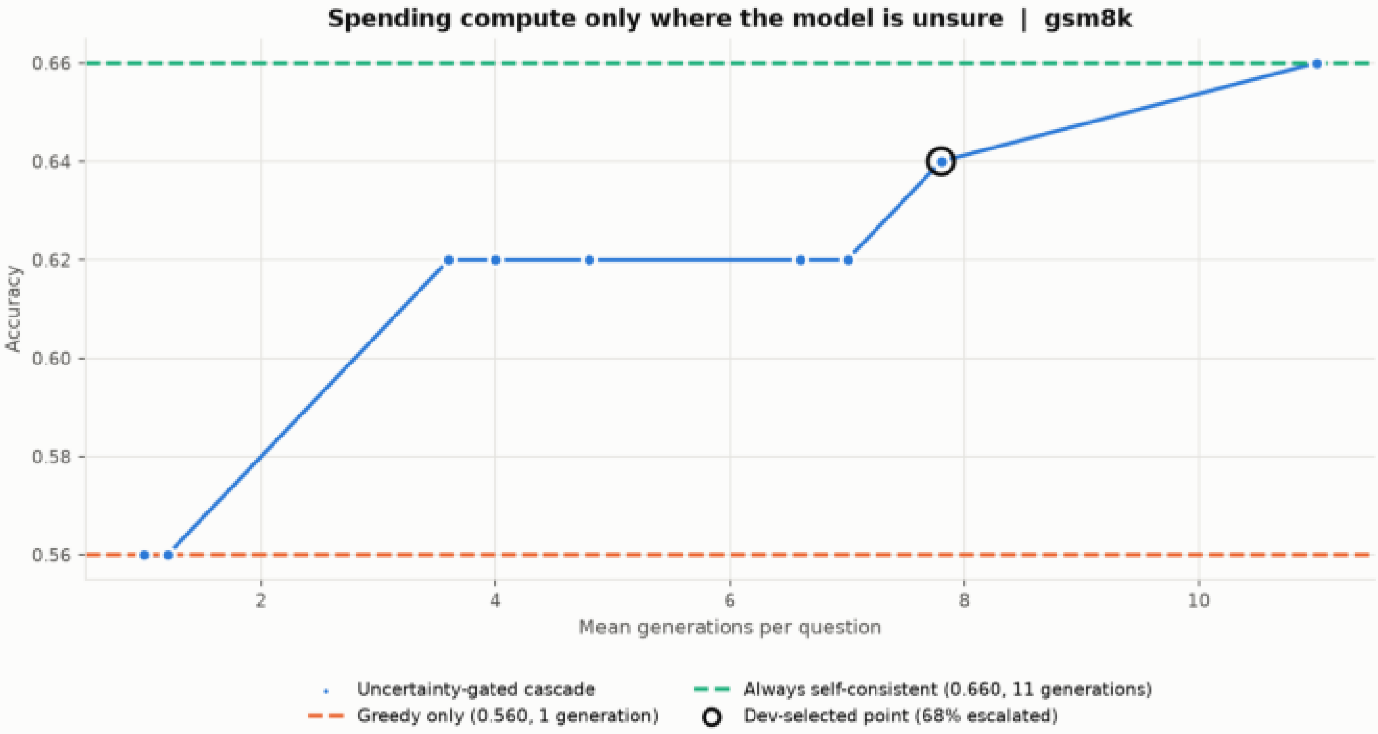
reliability_equal_mass_gsm8k.png

Calibration, equal-mass bins — nq_open



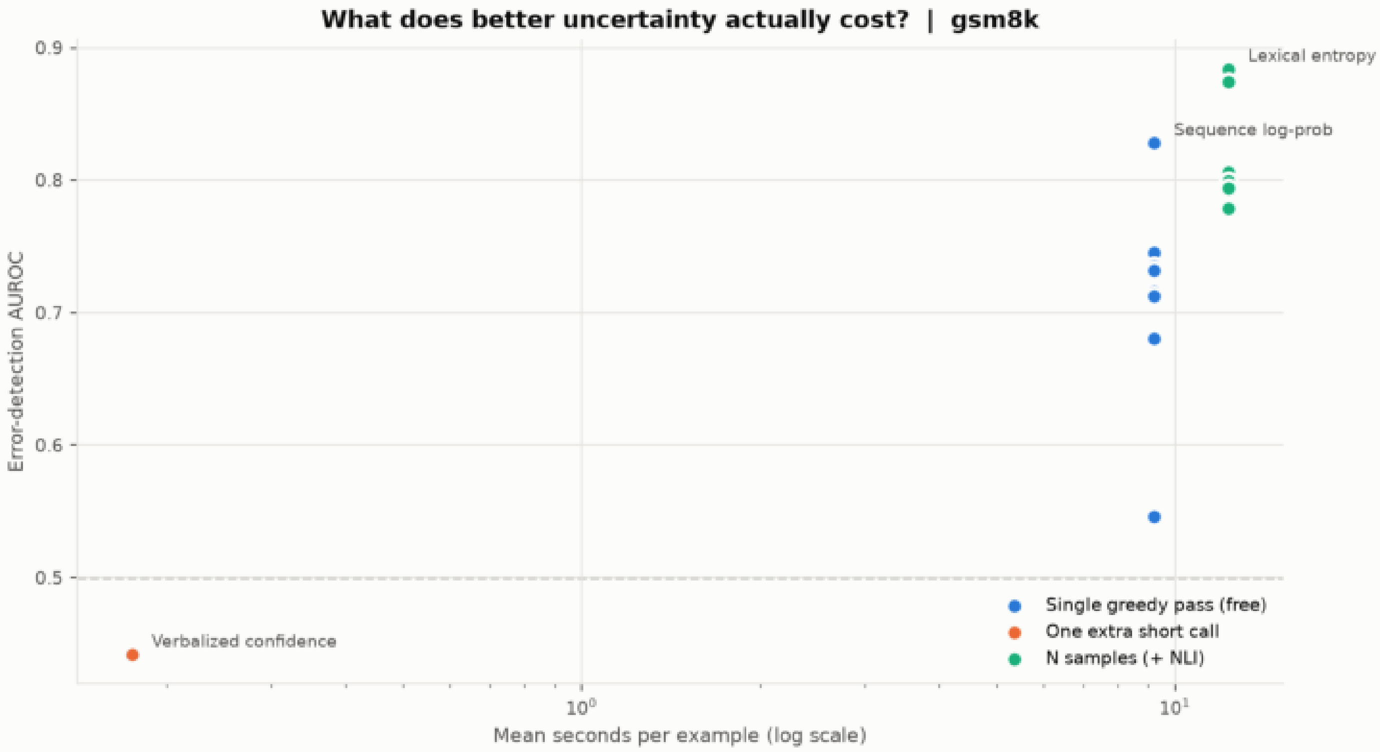
reliability_equal_mass_nq_open.png

Spending compute only where the model is unsure — gsm8k



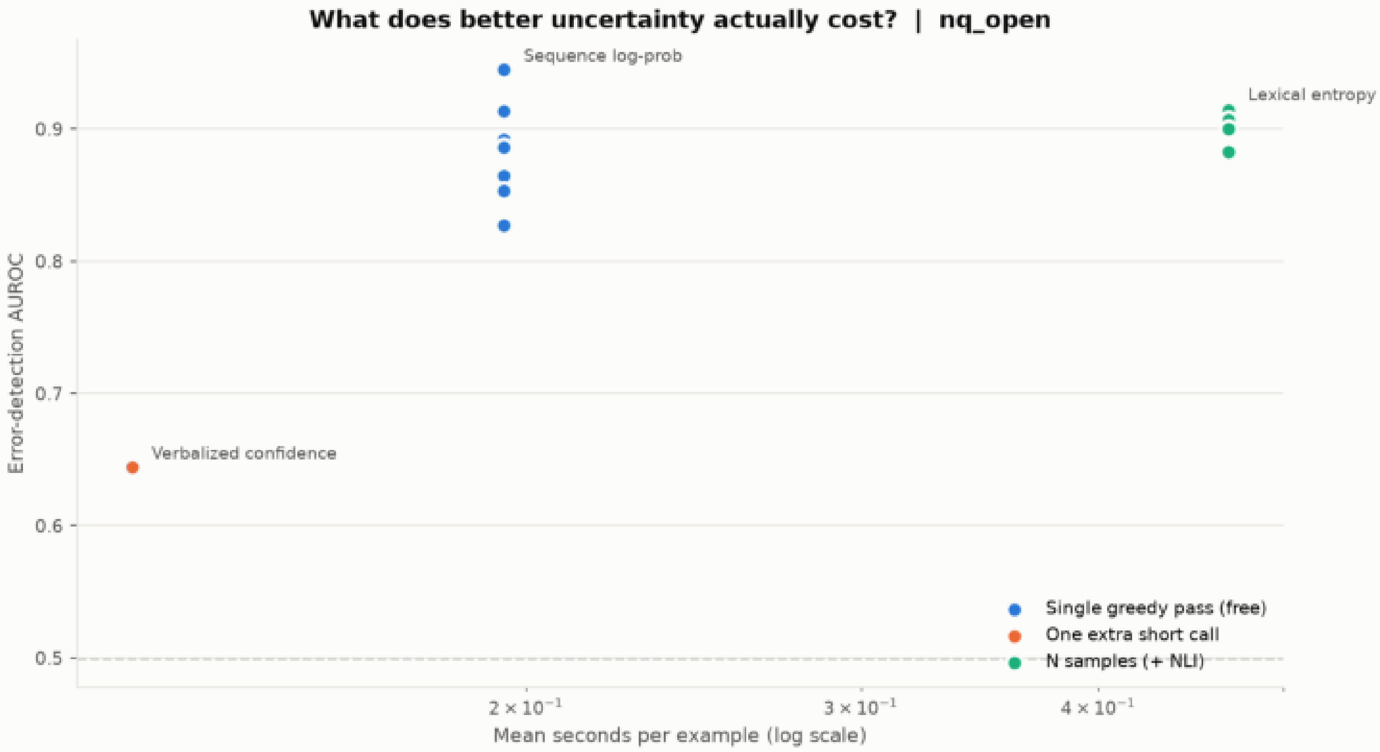
cascade_gsm8k.png

What better uncertainty actually costs — gsm8k



cost_quality_gsm8k.png

What better uncertainty actually costs — nq_open



cost_quality_nq_open.png